# S4_1 Lab Screenshots

## Sprint 4 Lab 1: Repository Pattern, pytest-mock, and CLI Retirement

**Student Name:** _____

**Date:** _____

**GitHub Repo URL:** _____

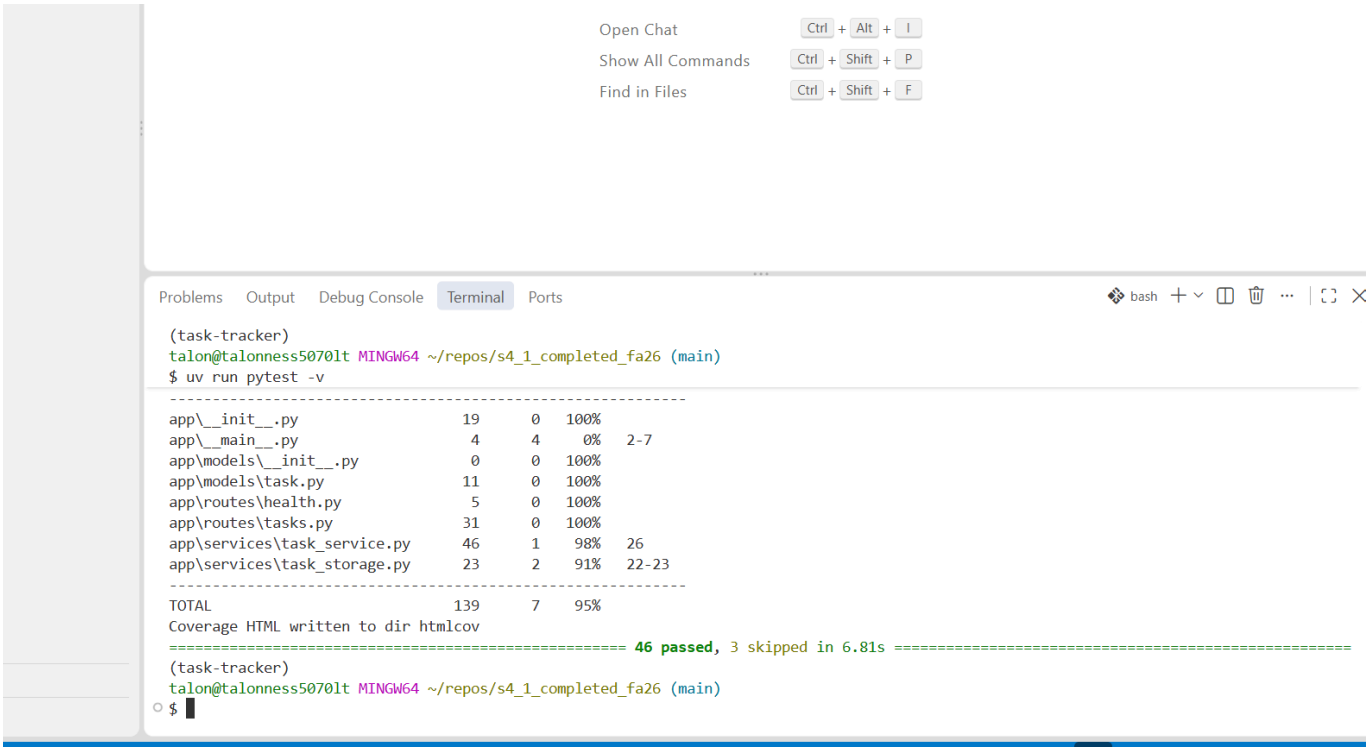
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> **Instructions:**
>
> 1. Save this file as `images/s4/lastname-S4_1-screenshots.md`
> 2. Replace each placeholder below with your actual screenshot
> 3. Export to PDF: right-click in VS Code → **Markdown PDF: Export (PDF)**
> 4. Rename: `lastname-S4_1-screenshots.pdf`
> 5. Submit to Blackboard **S4_1 Lab** assignment

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## Screenshot 1 — S3_2 Baseline (All Tests Passing)

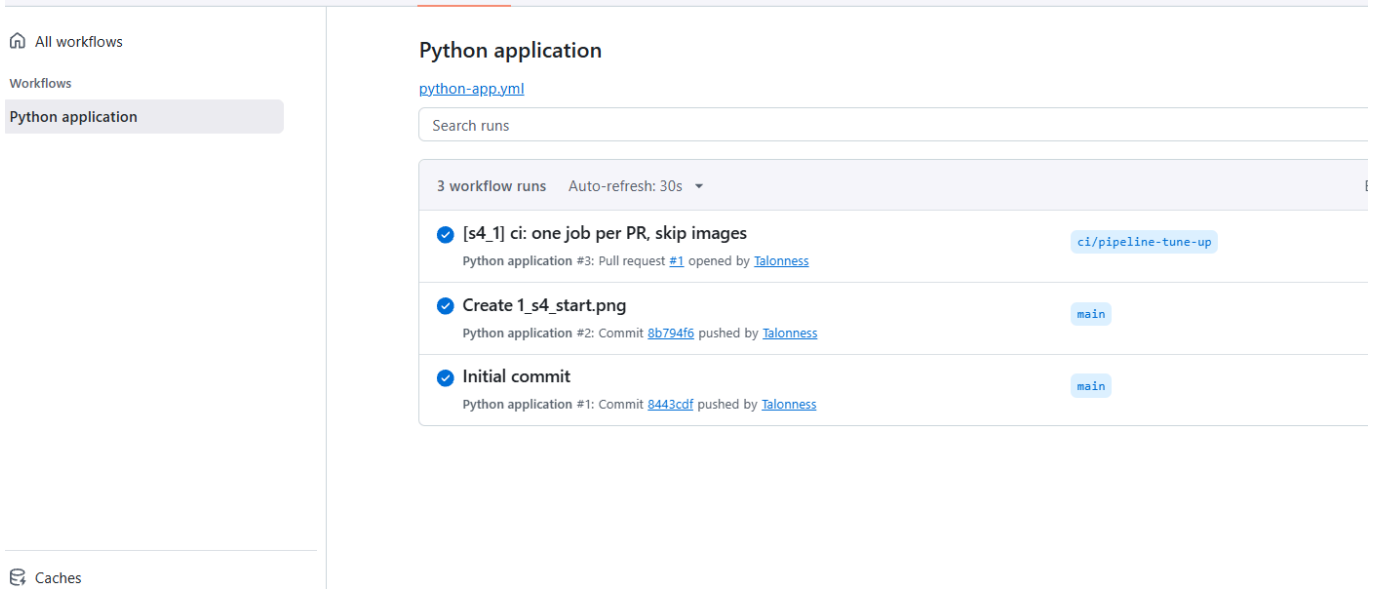
*Phase 1 Part 3 — Full test suite passing before S4_1 work begins. Server running in Terminal 1.*




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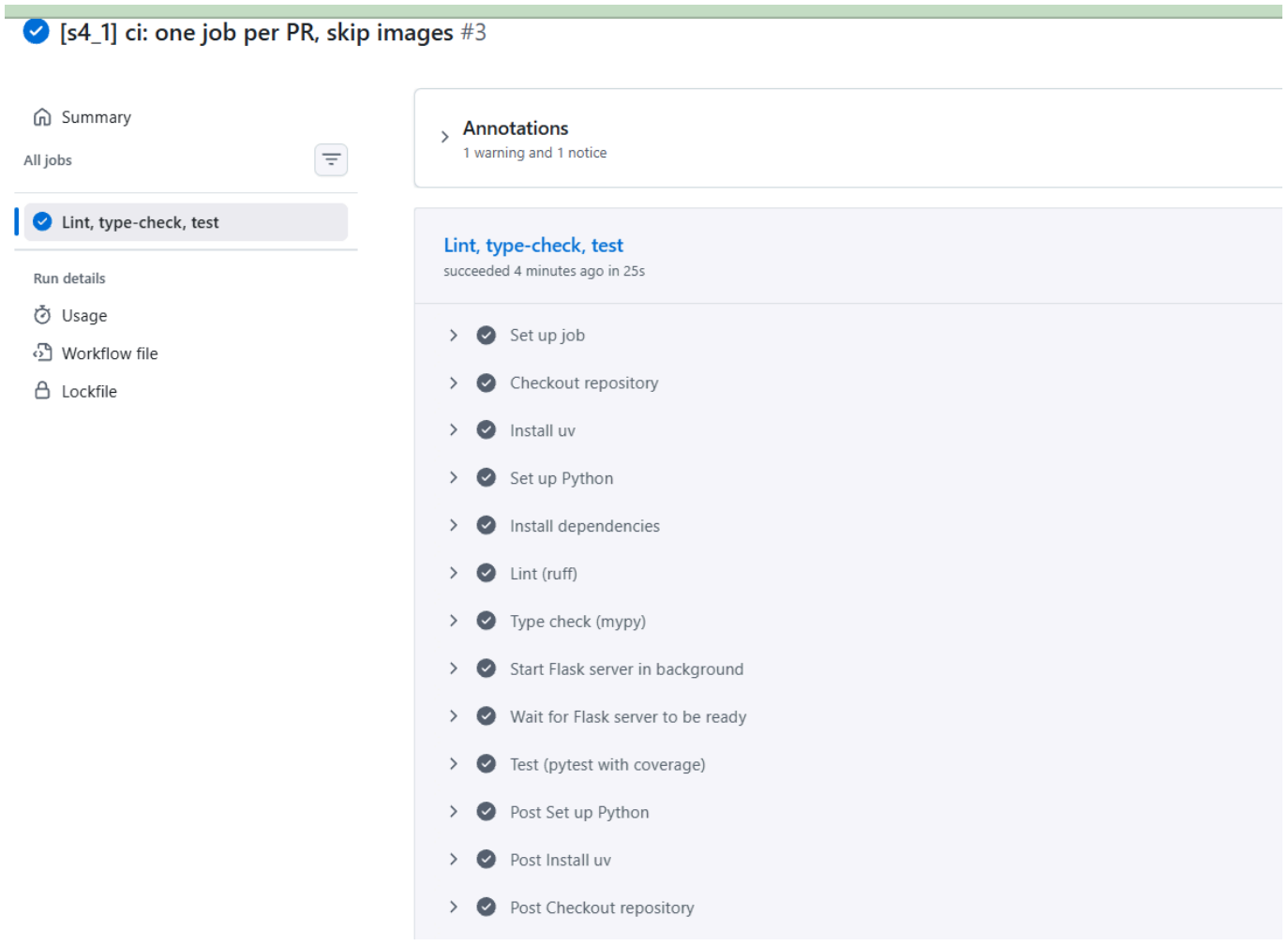
## Screenshot 1a — CI Runs Once Per PR

*Phase 1A Part 4 — Actions tab filtered to Python application, showing your `[s4_1] ci:` PR run at the top with no merge run after it.*



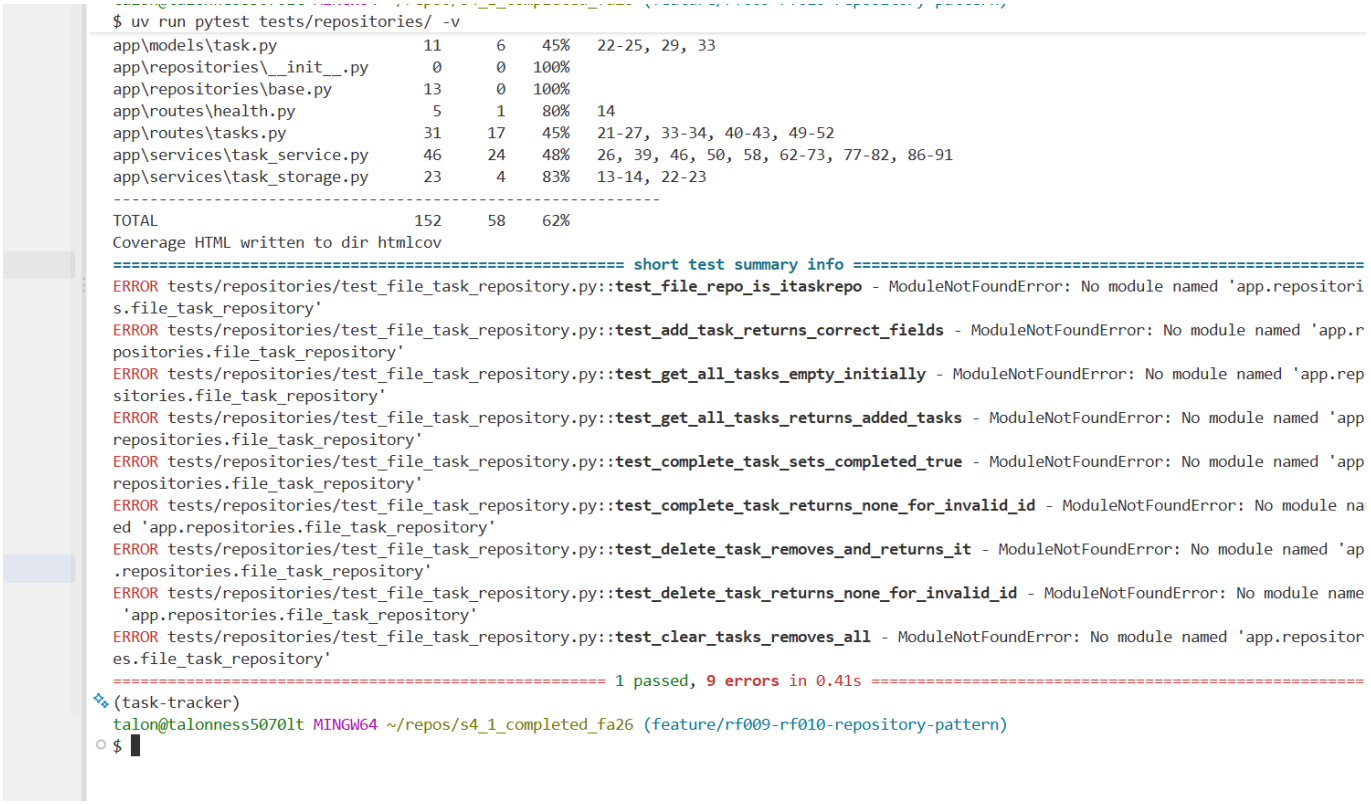

# Screenshot 1b — Single Job, Three Steps

*Phase 1A Part 4 — The single Lint, type-check, test job with its steps expanded and green.*




# Screenshot 2 — Contract Tests RED

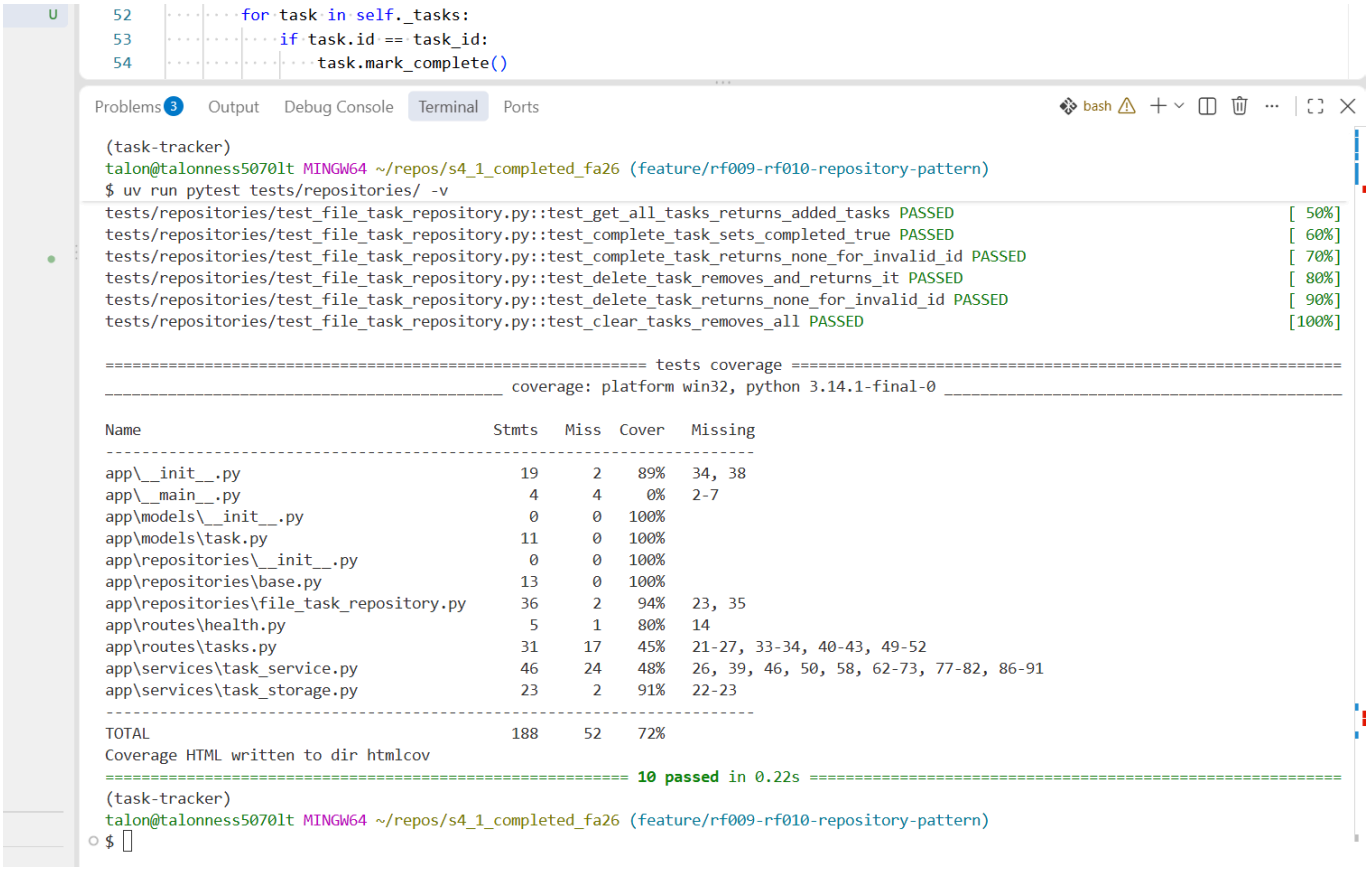
*Phase 2 Part 2 — Contract tests in ERROR state. FileTaskRepository not yet implemented.*



```
$ uv run pytest tests/repositories/ -v
app\models\task.py                11      6    45%   22-25, 29, 33
app\repositories\__init__.py       0      0   100%
app\repositories\base.py          13      0   100%
app\routes\health.py               5      1    80%   14
app\routes\tasks.py               31     17    45%   21-27, 33-34, 40-43, 49-52
app\services\task_service.py      46     24    48%   26, 39, 46, 50, 58, 62-73, 77-82, 86-91
app\services\task_storage.py      23      4    83%   13-14, 22-23
------------------------------------------------------------
TOTAL                            152     58    62%
Coverage HTML written to dir htmlcov
=========================================================== short test summary info ============================================================
ERROR tests/repositories/test_file_task_repository.py::test_file_repo_is_itaskrepo - ModuleNotFoundError: No module named 'app.repositori
s.file_task_repository'
ERROR tests/repositories/test_file_task_repository.py::test_add_task_returns_correct_fields - ModuleNotFoundError: No module named 'app.r
positories.file_task_repository'
ERROR tests/repositories/test_file_task_repository.py::test_get_all_tasks_empty_initially - ModuleNotFoundError: No module named 'app.rep
sitories.file_task_repository'
ERROR tests/repositories/test_file_task_repository.py::test_get_all_tasks_returns_added_tasks - ModuleNotFoundError: No module named 'app
repositories.file_task_repository'
ERROR tests/repositories/test_file_task_repository.py::test_complete_task_sets_completed_true - ModuleNotFoundError: No module named 'app
repositories.file_task_repository'
ERROR tests/repositories/test_file_task_repository.py::test_complete_task_returns_none_for_invalid_id - ModuleNotFoundError: No module na
ed 'app.repositories.file_task_repository'
ERROR tests/repositories/test_file_task_repository.py::test_delete_task_removes_and_returns_it - ModuleNotFoundError: No module named 'ap
.repositories.file_task_repository'
ERROR tests/repositories/test_file_task_repository.py::test_delete_task_returns_none_for_invalid_id - ModuleNotFoundError: No module name
 'app.repositories.file_task_repository'
ERROR tests/repositories/test_file_task_repository.py::test_clear_tasks_removes_all - ModuleNotFoundError: No module named 'app.repositor
es.file_task_repository'
========================================================== 1 passed, 9 errors in 0.41s ==========================================================
(task-tracker)
talon@talonness5070lt MINGW64 ~/repos/s4_1_completed_fa26 (feature/rf009-rf010-repository-pattern)
$
```

# Screenshot 3 — Contract Tests GREEN

*Phase 2 Part 3 — All FileTaskRepository contract tests passing after implementation.*



```
52          for task in self._tasks:
53              if task.id == task_id:
54                  task.mark_complete()

Problems 3   Output   Debug Console   Terminal   Ports                                                                    bash

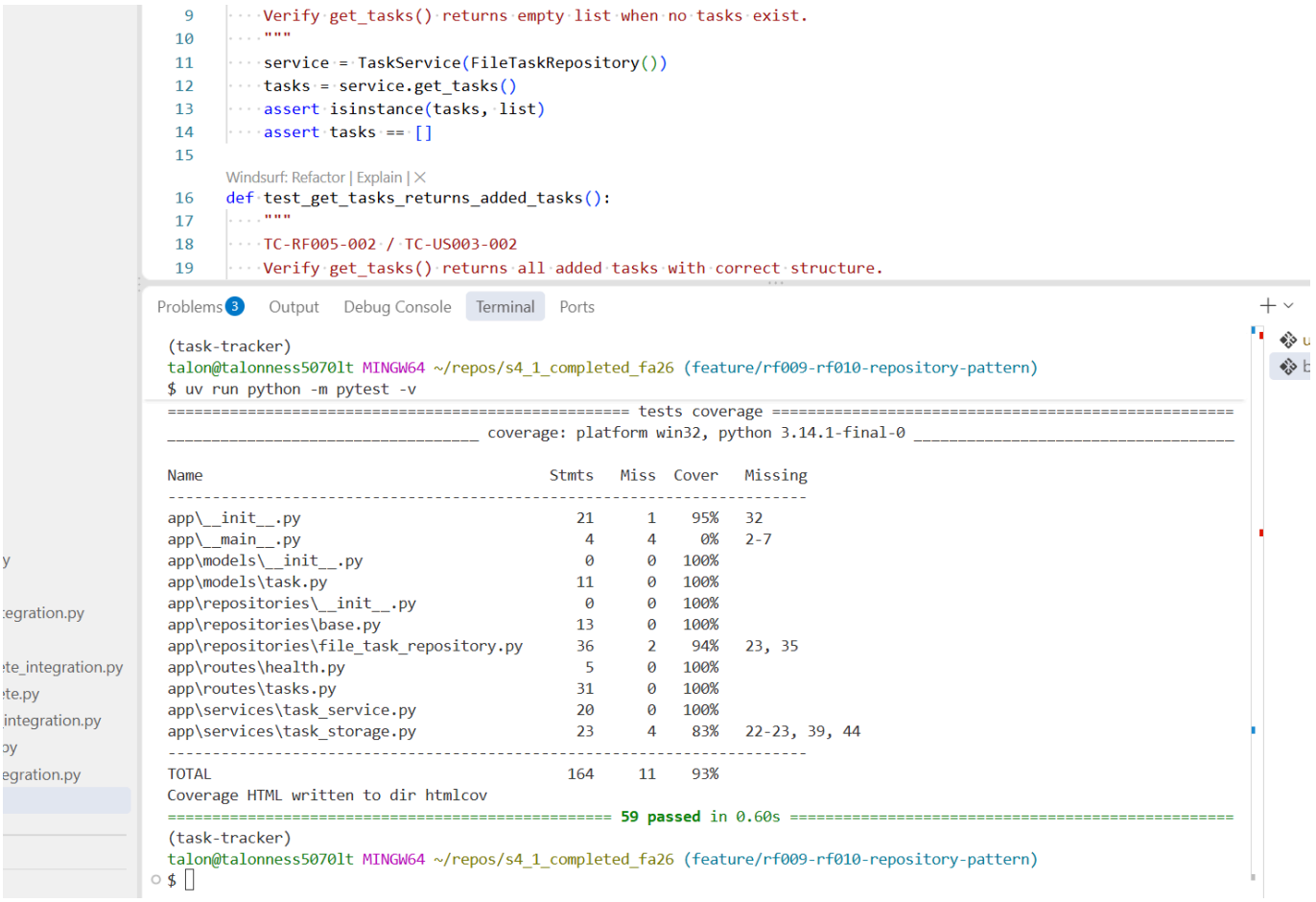
(task-tracker)
talon@talonness5070lt MINGW64 ~/repos/s4_1_completed_fa26 (feature/rf009-rf010-repository-pattern)
$ uv run pytest tests/repositories/ -v
tests/repositories/test_file_task_repository.py::test_get_all_tasks_returns_added_tasks PASSED                    [ 50%]
tests/repositories/test_file_task_repository.py::test_complete_task_sets_completed_true PASSED                    [ 60%]
tests/repositories/test_file_task_repository.py::test_complete_task_returns_none_for_invalid_id PASSED            [ 70%]
tests/repositories/test_file_task_repository.py::test_delete_task_removes_and_returns_it PASSED                   [ 80%]
tests/repositories/test_file_task_repository.py::test_delete_task_returns_none_for_invalid_id PASSED              [ 90%]
tests/repositories/test_file_task_repository.py::test_clear_tasks_removes_all PASSED                              [100%]

============================================================ tests coverage ============================================================
_____________________________________________ coverage: platform win32, python 3.14.1-final-0 _____________________________________________

Name                                       Stmts   Miss  Cover   Missing
------------------------------------------------------------------------
app\__init__.py                               19      2    89%   34, 38
app\__main__.py                                4      4     0%   2-7
app\models\__init__.py                         0      0   100%
app\models\task.py                            11      0   100%
app\repositories\__init__.py                   0      0   100%
app\repositories\base.py                      13      0   100%
app\repositories\file_task_repository.py      36      2    94%   23, 35
app\routes\health.py                           5      1    80%   14
app\routes\tasks.py                           31     17    45%   21-27, 33-34, 40-43, 49-52
app\services\task_service.py                  46     24    48%   26, 39, 46, 50, 58, 62-73, 77-82, 86-91
app\services\task_storage.py                  23      2    91%   22-23
------------------------------------------------------------------------
TOTAL                                        188     52    72%
Coverage HTML written to dir htmlcov
========================================================== 10 passed in 0.22s ==========================================================
(task-tracker)
talon@talonness5070lt MINGW64 ~/repos/s4_1_completed_fa26 (feature/rf009-rf010-repository-pattern)
$
```

# Screenshot 4 — Full Suite Passing After Repository Refactor

*Phase 2 Part 8 — All tests passing after TaskService, create_app(), and all 7 service test files updated.*



```
    Verify get_tasks() returns empty list when no tasks exist.
    """
    service = TaskService(FileTaskRepository())
    tasks = service.get_tasks()
    assert isinstance(tasks, list)
    assert tasks == []

Windsurf: Refactor | Explain |
def test_get_tasks_returns_added_tasks():
    """
    TC-RF005-002 / TC-US003-002
    Verify get_tasks() returns all added tasks with correct structure.
```

Problems 3 Output Debug Console Terminal Ports

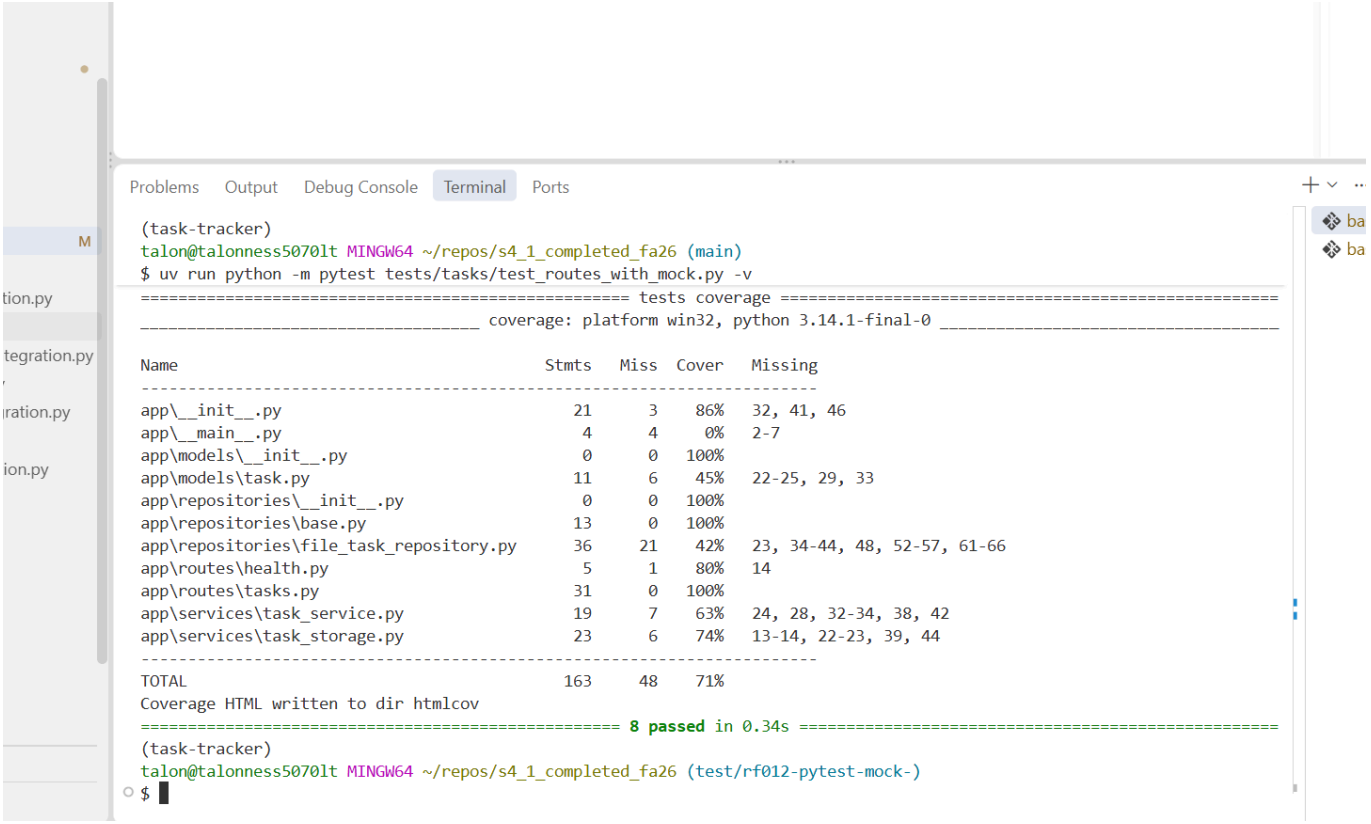
```
(task-tracker)
talon@talonness5070lt MINGW64 ~/repos/s4_1_completed_fa26 (feature/rf009-rf010-repository-pattern)
$ uv run python -m pytest -v
---------- tests coverage ----------
_____ coverage: platform win32, python 3.14.1-final-0 _____

Name                                      Stmts   Miss  Cover   Missing
-----------------------------------------------------------------------
app\__init__.py                              21      1    95%   32
app\__main__.py                               4      4     0%   2-7
app\models\__init__.py                        0      0   100%
app\models\task.py                           11      0   100%
app\repositories\__init__.py                  0      0   100%
app\repositories\base.py                     13      0   100%
app\repositories\file_task_repository.py     36      2    94%   23, 35
app\routes\health.py                          5      0   100%
app\routes\tasks.py                          31      0   100%
app\services\task_service.py                 20      0   100%
app\services\task_storage.py                 23      4    83%   22-23, 39, 44
-----------------------------------------------------------------------
TOTAL                                       164     11    93%
Coverage HTML written to dir htmlcov
---------- 59 passed in 0.60s ----------
(task-tracker)
talon@talonness5070lt MINGW64 ~/repos/s4_1_completed_fa26 (feature/rf009-rf010-repository-pattern)
$
```

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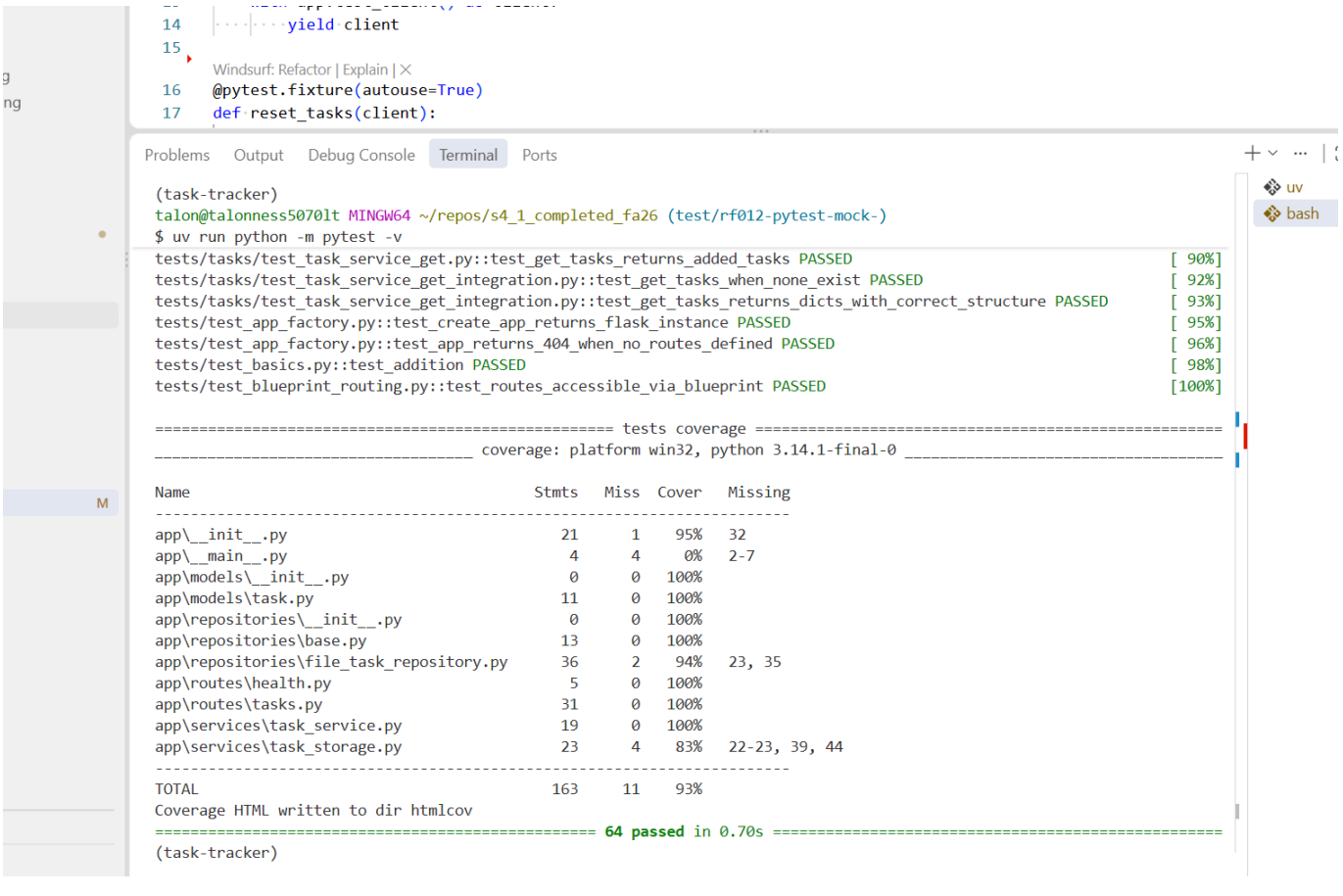
# Screenshot 5 — pytest-mock Route Tests Passing

*Phase 3 Part 1 — All 8 pytest-mock route tests passing in test_routes_with_mock.py.*




# Screenshot 6 — Full Suite After MockTaskService Removed

*Phase 3 Part 2 — All tests passing after MockTaskService removed from conftest.py.*



# Screenshot 7 — Full Suite After CLI Retired

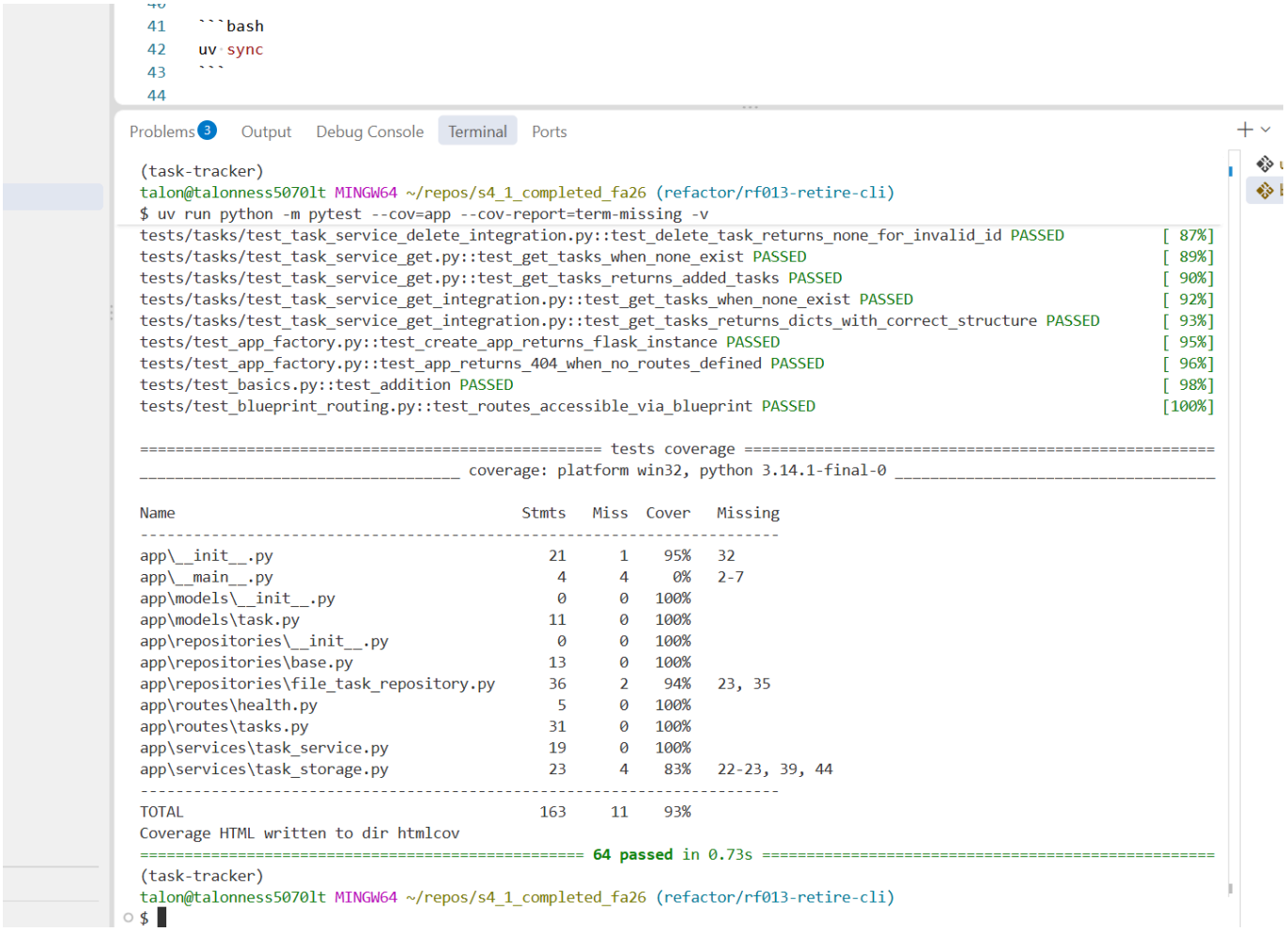
*Phase 4 Part 3 — All tests passing after cli/ directory removed.*



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# Screenshot 8 — Full Test Suite with Coverage Terminal

*Phase 5 Part 1 — Full test suite with `--cov` coverage report in terminal.*



```
41  ```bash
42  uv sync
43  ```
44
```

Problems 3 Output Debug Console Terminal Ports

```
(task-tracker)
talon@talonness5070lt MINGW64 ~/repos/s4_1_completed_fa26 (refactor/rf013-retire-cli)
$ uv run python -m pytest --cov=app --cov-report=term-missing -v
tests/tasks/test_task_service_delete_integration.py::test_delete_task_returns_none_for_invalid_id PASSED          [ 87%]
tests/tasks/test_task_service_get.py::test_get_tasks_when_none_exist PASSED                                      [ 89%]
tests/tasks/test_task_service_get.py::test_get_tasks_returns_added_tasks PASSED                                  [ 90%]
tests/tasks/test_task_service_get_integration.py::test_get_tasks_when_none_exist PASSED                          [ 92%]
tests/tasks/test_task_service_get_integration.py::test_get_tasks_returns_dicts_with_correct_structure PASSED     [ 93%]
tests/test_app_factory.py::test_create_app_returns_flask_instance PASSED                                         [ 95%]
tests/test_app_factory.py::test_app_returns_404_when_no_routes_defined PASSED                                    [ 96%]
tests/test_basics.py::test_addition PASSED                                                                       [ 98%]
tests/test_blueprint_routing.py::test_routes_accessible_via_blueprint PASSED                                     [100%]

---------- coverage: platform win32, python 3.14.1-final-0 ----------

Name                                      Stmts   Miss  Cover   Missing
-----------------------------------------------------------------------
app\__init__.py                              21      1    95%   32
app\__main__.py                               4      4     0%   2-7
app\models\__init__.py                        0      0   100%
app\models\task.py                           11      0   100%
app\repositories\__init__.py                  0      0   100%
app\repositories\base.py                     13      0   100%
app\repositories\file_task_repository.py     36      2    94%   23, 35
app\routes\health.py                          5      0   100%
app\routes\tasks.py                          31      0   100%
app\services\task_service.py                 19      0   100%
app\services\task_storage.py                 23      4    83%   22-23, 39, 44
-----------------------------------------------------------------------
TOTAL                                       163     11    93%
Coverage HTML written to dir htmlcov
================================================== 64 passed in 0.73s ==================================================
(task-tracker)
talon@talonness5070lt MINGW64 ~/repos/s4_1_completed_fa26 (refactor/rf013-retire-cli)
$
```

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# Screenshot 9 — HTML Coverage Report

*Phase 5 Part 2 — HTML coverage report showing file breakdown in browser.*

Coverage report: 93%

Files Functions Classes

filter...

hide coverec

*coverage.py v7.14.2, created at 2026-10-18 19:57 -0400*

| File ▲ | statements | missing | excluded | coverage |
|---|---|---|---|---|
| app__init__.py | 21 | 1 | 0 | 95% |
| app__main__.py | 4 | 4 | 0 | 0% |
| app\models__init__.py | 0 | 0 | 0 | 100% |
| app\models\task.py | 11 | 0 | 0 | 100% |
| app\repositories__init__.py | 0 | 0 | 0 | 100% |
| app\repositories\base.py | 13 | 0 | 10 | 100% |
| app\repositories\file_task_repository.py | 36 | 2 | 0 | 94% |
| app\routes\health.py | 5 | 0 | 0 | 100% |
| app\routes\tasks.py | 31 | 0 | 0 | 100% |
| app\services\task_service.py | 19 | 0 | 0 | 100% |
| app\services\task_storage.py | 23 | 4 | 0 | 83% |
| **Total** | **163** | **11** | **10** | **93%** |

*coverage.py v7.14.2, created at 2026-10-18 19:57 -0400*

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*End of S4_1 Lab Screenshots*